WOXSEN UNIVERSITY

School of Technology

CAPSTONE PROJECT-I

(GUIDELINES)

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1 INTRODUCTION

The Capstone Project is a cornerstone of the B.Tech curriculum at Woxsen University's School of Technology, representing the culmination of four years of academic learning, innovation, and practical exposure. It serves as an integrative platform where students apply multidisciplinary knowledge to address real-world challenges through engineering design, research, and problem-solving.

This project bridges the gap between theoretical concepts learned in classrooms and their practical implementation in industry or research contexts. Students are encouraged to identify pressing societal or industrial problems, conceptualize feasible solutions, and develop functional prototypes or software applications that demonstrate measurable outcomes.

The Capstone Project not only assesses technical proficiency but also enhances critical thinking, teamwork, communication, and project management skills—core competencies essential for professional success. It challenges students to engage in innovation, explore emerging technologies such as Artificial Intelligence, IoT, Cloud Computing, or Sustainable Engineering, and apply them creatively to deliver impactful solutions.

Furthermore, the project fosters an entrepreneurial mindset by motivating students to convert their ideas into viable products or startups. Through continuous mentorship, peer collaboration, and formal evaluations, the Capstone Project nurtures professional ethics, accountability, and leadership, making students industry-ready and research-oriented upon graduation.

2 PROJECT DETATIL

The Capstone Project–I is a vital component of the Bachelor of Technology (B.Tech) curriculum at Woxsen University's School of Technology, undertaken during the 7th Semester of the program. Spanning six months (one full semester) and carrying 4 academic credits, this course emphasizes the practical implementation of engineering knowledge through innovation, design thinking, and multidisciplinary collaboration.

This project provides students with the opportunity to integrate theoretical understanding acquired throughout their coursework and apply it to a tangible, real-world problem. The focus is on developing and validating working prototypes or systems that demonstrate measurable technical, industrial, or societal value. Students are encouraged to engage in applied research, experimentation, and design to showcase originality, feasibility, and technical rigor in their work.

The Capstone Project–I acts as the foundation for Capstone Project–II, undertaken in the 8th semester, where the outcomes of this phase are further refined, validated, and extended toward final deployment or publication.

Table 1: Program Details

Program	**Bachelor of Technology (B.Tech)**
Semester	**7th Semester**
Duration	**6 Months (One Semester)**
Credits	**4 Credits**
Project Type	**Capstone Project-I (Practical Implementation Focus)**

2.1 Course Outcomes (COs)

Upon successful completion of **Capstone Project–I**, students will be equipped with a comprehensive set of skills that reflect their readiness for professional engineering practice and

research. They will be able to identify and critically analyze real-world technical or societal challenges, defining precise problem statements aligned with SMART objectives (CO1). Through the application of suitable tools, technologies, and methodologies, students will design and implement innovative solutions that effectively address identified gaps or limitations in existing systems (CO2). They will further demonstrate their technical proficiency and creativity by developing functional prototypes or systems that exhibit novelty and potential impact in industrial or social contexts (CO3). Finally, students will develop the ability to document, present, and communicate their project findings through well-structured technical reports and professional presentations, adhering to institutional and academic standards of quality (CO4).

Table 2: Course Outcomes (COs) of Capstone Project–I

CO Number	Course Outcome
CO1	Identify and analyze real-world technical or societal problems, formulating clear problem statements with measurable objectives aligned with SMART criteria.
CO2	Design and implement innovative solutions using appropriate tools, technologies, and methodologies to address identified gaps in existing systems.
CO3	Develop functional prototypes or systems demonstrating technical proficiency, novelty, and potential for industry or societal impact.
CO4	Document and communicate project outcomes effectively through technical reports and presentations adhering to academic and professional standards.

Collectively, these outcomes ensure that students integrate theoretical knowledge with practical problem-solving, innovation, and communication skills essential for engineering excellence.

2.2 CO-PO Mapping

The following table illustrates the mapping between Course Outcomes (COs) and Program Outcomes (POs). The correlation levels are indicated as: **1 = Low**, **2 = Medium**, **3 = High**.

Table 3 : CO–PO Mapping for Capstone Project–I

CO	Program Outcomes (POs)											
	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO9**	**PO10**	**PO11**	**PO12**
CO 1	**3**	**3**	**2**	**2**	**2**	**1**	**1**	**2**	**2**	**2**	**1**	**3**
CO 2	**3**	**3**	**3**	**3**	**3**	**2**	**2**	**2**	**2**	**2**	**2**	**3**
CO 3	**3**	**3**	**3**	**2**	**3**	**3**	**2**	**2**	**2**	**3**	**2**	**3**
CO 4	**2**	**2**	**2**	**1**	**2**	**2**	**1**	**2**	**3**	**3**	**3**	**3**

Note: *PO1-PO12 represent the standard B.Tech Program Outcomes covering Engineering Knowledge, Problem Analysis, Design/Development of Solutions, Conduct Investigations, Modern Tool Usage, Engineer and Society, Environment and Sustainability, Ethics, Individual and Team Work, Communication, Project Management and Finance, and Life-long Learning.*

As shown in **Table 2.2**, the Course Outcomes (COs) of **Capstone Project–I** are systematically mapped to the twelve **Program Outcomes (POs)** defined for the B.Tech curriculum. Each numerical value represents the degree of correlation, where **1 denotes a low correlation**, **2 indicates a medium correlation**, and **3 signifies a high correlation** between a specific CO and PO.

The mapping highlights that all course outcomes strongly contribute to core program competencies such as **engineering knowledge (PO1)**, **problem analysis (PO2)**, and **design and development of solutions (PO3)**. A consistently high correlation (level 3) across multiple outcomes also reflects the project's emphasis on **innovation, technical proficiency, communication, and life-long learning (PO12)**. This alignment ensures that the Capstone Project–I not only develops technical expertise but also promotes holistic professional attributes, aligning with national accreditation standards such as NBA and NAAC for outcome-based education.

3 REPORT GUIDELINES

The project report must adhere to the following structured format to ensure clarity, professionalism, and compliance with academic standards. Each section serves a specific purpose in documenting your project comprehensively.

3.1 Title Page

The title page must include the following elements in the specified order:

- Project Title: **A clear, concise, and descriptive title reflecting the project's purpose and scope. The title should be specific enough to convey the project's focus without being overly lengthy.**
- Institute Logo: **Include the official logo of Woxsen University at the top center of the page.**
- Name(s) of Student(s) with Roll No(s).: **List all team members' full names along with their student identification numbers. Ensure names are spelled correctly and IDs are accurate.**
- Guide/Faculty Name: **Provide the full name and designation of the assigned faculty mentor/supervisor.**
- Department and Institute Name: **Specify the department (e.g., Department of Computer Science and Engineering) and "Woxsen University, School of Technology".**
- Academic Year & Semester: **Mention the academic year (e.g., 2025–2026) and specify "7th Semester".**

Example Format:

Development of IoT-Based Smart Irrigation System for Precision Agriculture
Submitted by: John Doe (12345678), Jane Smith (12345679)
Under the guidance of: Dr. A. Kumar, Associate Professor.

3.2 Index (Table of Contents)

- List all sections and subsections with corresponding page numbers.
- Include headings for Abstract, Introduction, Technology Review, Methodology, Results, Conclusion, References, etc.
- Use consistent formatting with decimal numbering (e.g., 1, 1.1, 1.2, 1.2.1).
- Ensure page numbers are accurate and updated before final submission.

- The table of contents should be automatically generated using word processing software features for accuracy.

3.3 Abstract

Word Limit: 150–200 words

Content Requirements:

- Purpose: **Briefly state the project's aim and the specific problem being addressed. Clearly articulate why this project was undertaken.**
- Methodology/Approach: **Summarize the key tools, technologies, or methods employed in the project implementation. Mention the primary technical approach.**
- Key Outcomes/Results: **Highlight major achievements, deliverables, or findings. Include quantifiable results where possible.**
- Industry Relevance/Societal Impact: **Emphasize the practical applications, benefits to industry or society, and potential impact of the project.**

Important: The abstract should be written last, after completing the entire report, as it summarizes the complete project. It should be self-contained and understandable without referring to other sections.

3.4 Introduction

Content Requirements:

Problem Statement: Clearly define the technical or real-world problem being addressed. Explain the current challenges and why existing solutions are inadequate. Use specific examples and data where available.

- Background: Provide comprehensive context including:
  - Relevant domain knowledge
  - Current industry trends and practices
  - Historical context of the problem
  - Technological evolution in the related field
- Significance and Scope: Explain the project's importance by addressing:
  - Who will benefit from this project
  - What impact it will have
  - Technical boundaries (what is included/excluded)
  - Geographical or application-specific limitations

- Time frame considerations

3.5 Technology Review

Conduct a comprehensive review of existing technologies, tools, frameworks, and systems relevant to your project. Present your findings in a structured table format with the following columns:

Column Name	Description
Technology Name	The name or identifier of the tool, framework, platform, or system (e.g., TensorFlow, Arduino, AWS, React Native).
Description	A brief overview of the technology's purpose and primary functionality. Explain what it does and its intended use cases.
Key Features	Core functionalities or capabilities relevant to the project's requirements (e.g., real-time processing, IoT support, machine learning capabilities).
Performance	Metrics or qualitative assessment of efficiency, speed, or reliability (e.g., latency, throughput, uptime, processing speed).
Scalability	Ability to handle increased loads or scale with project growth (e.g., supports 100 to 10,000 users, horizontal/vertical scaling capabilities).
Cost/Licensing	Pricing model (e.g., free, subscription-based, one-time purchase) and licensing terms (e.g., open-source, proprietary, MIT License, GPL).
Compatibility	Integration capability with existing systems, hardware, or software used in the project. List supported platforms and dependencies.

Ease of Use	User-friendliness, learning curve, availability of documentation, tutorials, and community support. Rate based on developer experience.
Security	Security features, compliance with standards (e.g., ISO, GDPR), encryption capabilities, and known vulnerabilities.
Current Market	Market adoption, popularity, commercial trends (e.g., market share, number of active users, industry adoption rate, growth trends).
Limitations	Key drawbacks or constraints (e.g., high resource usage, limited platform support, vendor lock-in, performance bottlenecks).
Relevance to Project	Explain how well the technology aligns with the project's objectives and requirements. Justify selection or rejection.

Note: Review at least 5-8 technologies relevant to your project domain. Compare similar technologies to justify your final selection.

3.6 Review Summary

Content Requirements:

- Key Findings: Synthesize and summarize the most important findings from the technology review. Identify patterns, trends, and common features across reviewed technologies.
- Gaps and Limitations: Highlight specific gaps, limitations, or inefficiencies identified in existing systems and approaches. Be specific about what is missing or inadequate.
- Innovation Opportunities: Identify clear opportunities for innovation or improvement that your project will address. Connect these opportunities directly to the gaps identified.
- Justification for Project: Explain how your project will fill the identified gaps and improve upon existing solutions.

3.7 Limitations or Research Gap

Content Requirements:

- Clear Articulation: Provide a detailed explanation of limitations in existing technologies or research gaps identified during the review process.
- Justification: Justify the need for your proposed project by establishing a direct link between identified gaps and your project objectives.
- Evidence-Based: Support all claims with evidence from the technology review. Reference specific technologies, studies, or market data.
- Categorization: Organize gaps into categories such as:
    - Technical limitations (performance, scalability, compatibility)
    - Functional gaps (missing features or capabilities)
    - Usability issues (complexity, learning curve)
    - Cost-effectiveness concerns
    - Security or privacy shortcomings

3.8 Objectives

Content Requirements:

List specific, measurable, achievable, relevant, and time-bound (SMART) goals for your project. Each objective should:

- Specific: Clearly state what you will accomplish
- Measurable: Include quantifiable metrics or deliverables
- Achievable: Be realistic given time and resource constraints
- Relevant: Address technical challenges and align with industry or societal needs
- Time-bound: Specify completion timeline within the semester

Example Objectives:

1. Develop a low-cost IoT-based monitoring system to improve agricultural yield by 20% within 6 months.
2. Design and implement a machine learning algorithm achieving 90% accuracy in fault detection by Week.
3. Create a functional mobile application prototype with real-time data visualization capabilities by the end of the semester.

Number of Objectives: Typically 3-5 objectives are appropriate for a semester-long project.

3.9 Methodology

Content Requirements:

- Approach: Provide a detailed explanation of the implementation process including:
    - System design and architecture
    - Development phases (planning, design, implementation, testing)
    - Testing and validation strategies
    - Quality assurance procedures
- Tools and Technologies: Specify all resources used:
    - Programming languages (e.g., Python, Java, JavaScript)
    - Frameworks and libraries (e.g., TensorFlow, React, Spring Boot)
    - Hardware components (e.g., Arduino, Raspberry Pi, sensors)
    - Software tools (e.g., IDEs, databases, cloud platforms)
    - Development methodologies (e.g., Agile, Waterfall)
- Block Diagram: Include a clear, labeled, and numbered block diagram illustrating:
    - System architecture
    - Component interactions
    - Data flow
    - User interfaces
- Project Timeline and Milestones: Provide a Gantt chart or detailed table outlining:
    - Key project phases with start and end dates
    - Major milestones and deliverables
    - Dependencies between tasks
    - Resource allocation

Format: Use diagrams, flowcharts, algorithms, or pseudocode to enhance clarity and understanding. All figures must be properly labeled and referenced in the text.

3.10 Novelty

This section is critical for demonstrating the innovative aspects of your project. Address the following:

- Unique Contribution: Clearly highlight the project's novel feature or approach. This could be:
    - A new algorithm or optimization technique
    - Innovative system design or architecture
    - Novel application of existing technology
    - Unique combination of multiple technologies
    - Improved methodology or process
- Gap Addressing: Link the novelty directly to specific limitations identified in the Technology Review. Explain how your innovation addresses these gaps effectively.
- Technical Advancement: Describe how the project improves upon existing methods or tools. Provide specific comparisons:
    - Performance improvements (speed, accuracy, efficiency)
    - Cost reduction
    - Enhanced usability
    - Better scalability
    - Improved security
- Industry/Societal Impact: Explain the practical or societal benefits of the novel aspect:
    - How it solves real-world challenges
    - Potential market applications
    - Social or environmental benefits
    - Economic advantages
- Evidence of Novelty: Provide preliminary results, comparisons with existing solutions, or proof-of-concept demonstrations that support your innovation claims. Reference the Technology Review to show how your approach differs.

3.11 Results/Outcome

Content Requirements:

- Achieved Outcomes: Present all outcomes achieved during the project, including:
    - Functional prototypes or systems developed
    - Test results and performance metrics
    - User testing feedback
    - Validation of objectives
- Data and Evidence: Support all claims with:
    - Quantitative data (tables, statistics, measurements)
    - Screenshots of working systems
    - Simulation results
    - Performance graphs and charts
    - Comparison data with existing solutions
- Technical Achievements: Highlight specific technical accomplishments:
    - Successfully implemented features
    - Performance benchmarks achieved
    - Problems solved
    - Innovations demonstrated
- Practical Implications: Discuss real-world applications and benefits:
    - Industry applications
    - Societal impact
    - Scalability potential
    - Commercial viability

Format: Include labeled and numbered figures, tables, and graphs (e.g., "Figure 1: System Performance Comparison", "Table 2: Test Results Summary"). Ensure all visual elements are clear, high-quality, and properly referenced in the text.

3.12 Conclusion

Content Requirements:

- **Summary of Key Findings:** Concisely summarize the major outcomes and achievements of the project. Restate how objectives were met.
- **Significance:** Discuss the importance and impact of your findings:
    - Contribution to the field
    - Problem-solving effectiveness
    - Value addition to existing knowledge or practice
- **Limitations:** Honestly discuss limitations of the current work:
    - Technical constraints (scalability, performance)
    - Resource limitations (time, budget, hardware)
    - Scope restrictions
    - Known issues or bugs
    - Areas requiring further development
- **Lessons Learned:** Reflect on the project experience:
    - Technical challenges faced and overcome
    - Teamwork and collaboration insights
    - Project management learnings
    - Skills developed
    - What would be done differently

3.13 Recommendation and Future Scope

Content Requirements:

- Improvements for Capstone Project-II: Identify specific opportunities for enhancement in the 8th semester:
    - Features to be added
    - Performance optimizations
    - Extended functionality
    - Enhanced user experience
- **Research and Industrial Applications:** Discuss potential directions:

    - Scaling the solution for commercial deployment
    - Integration with emerging technologies (AI, IoT, blockchain)
    - Adaptation for different domains or industries
    - Collaboration opportunities with industry partners
- **Long-term Vision:** Articulate the broader potential impact and evolution of the project beyond the academic setting.

3.14 Acknowledgment

Content Requirements:

- Acknowledge contributions from:
    - Faculty mentor/supervisor for guidance and support
    - Institute resources (laboratories, equipment, library)
    - Peers who provided assistance or feedback
    - External stakeholders (industry partners, experts)
    - Open-source communities or online resources utilized
    - Family members for support and encouragement
- Keep the acknowledgment concise (150-200 words), formal, and sincere.
- Maintain a professional tone throughout.

3.15 References

Content Requirements:

- Cite all sources used in the report including:
    - Academic journals and research papers
    - Conference proceedings
    - Books and textbooks
    - Technical documentation
    - Websites and online resources
    - Industry reports and white papers

- Format: Follow IEEE citation format consistently throughout. Examples:
    - Journal Article: [1] J. Doe and A. Smith, "Title of the Article," *Journal Name*, vol. 10, no. 3, pp. 45–50, Mar. 2023.
    - Conference Paper: [2] M. Johnson, "Conference Paper Title," in *Proc. IEEE Int. Conf. Technology*, New York, NY, USA, 2023, pp. 123-128.
    - Book: [3] R. Brown, *Book Title*, 2nd ed. New York, NY, USA: Publisher, 2022.
    - Website: [4] "Web Page Title," Website Name. [Online]. Available: https://www.example.com. [Accessed: Jan. 15, 2025].
- Ensure all in-text citations correspond to references in the bibliography.
- Use credible and recent sources (preferably within the last 5-10 years).
- Minimum of 15-20 references are expected for a comprehensive project.

3.16 Word Limit

- Total Word Count: **3,000–4,000 words** (excluding references, appendices, and diagrams)
- Plagiarism Requirements:
    - Reports must pass plagiarism check using institutional tools (e.g., Turnitin)
    - Maximum similarity index allowed: <15%
    - Direct quotations must be properly cited and minimal
    - Paraphrasing must be substantial and in your own words
    - Non-compliance will result in project disqualification
- Content Distribution Guidelines:
    - Introduction: 10-12%
    - Technology Review: 20-25%
    - Methodology: 20-25%
    - Results and Discussion: 25-30%
    - Conclusion and Future Scope: 10-12%

3.17 Formatting Guidelines

Typography and Layout:

- Font: Times New Roman, 12-point for body text
- Line Spacing: 1.5 throughout the document
- Margins: 1 inch (2.54 cm) on all sides
- Alignment: Justified for body text
- Page Numbers: Include on every page (bottom center or right), starting from the Introduction

Heading Styles:

- Heading 1 (Main Sections): Bold, 14-point, numbered (e.g., 1. Introduction)
- Heading 2 (Subsections): Bold, 13-point, numbered (e.g., 1.1 Background)
- Heading 3 (Sub-subsections): Bold, 12-point, numbered (e.g., 1.1.1 Problem Context)

Figures, Tables, and Diagrams:

- **All figures and tables must be:**
    - Clear and high-resolution (minimum 300 DPI for images)
    - Properly labeled with captions (e.g., "Figure 1: System Architecture")
    - Numbered sequentially
    - Referenced in the text before they appear
    - Centered on the page
- **Captions should be:**
    - Below figures (10-point font)
    - Above tables (10-point font)
    - Descriptive and self-explanatory

Writing Style:

- Maintain a formal academic tone throughout.
- Avoid colloquial language, slang, or casual expressions.
- Use third person (avoid "I", "we", "our").
- Be concise and precise.

- Use active voice where appropriate.
- Ensure grammatical correctness and proper punctuation.
- Define technical terms and acronyms on first use.

4 TEAM SELECTION AND TOPIC APPROVAL

4.1 Team Selection

- Team Size: 1 to 3 members per team
    - Solo projects allowed for highly motivated individuals
    - Teams of 2-3 members encouraged for collaborative learning
    - Teams larger than 3 members not permitted
- Team Formation:
    - Teams must be formed at the start of the 7th semester (Week 1-2)
    - Students may form teams based on common interests or assigned by faculty
    - Team composition form must be submitted for approval
- Team Continuity:
    - Teams remain consistent across both semesters (7th and 8th)
    - This ensures project continuity and deeper development
    - Team changes allowed only under exceptional circumstances with faculty approval
- Team Dynamics:
    - Members should have complementary skills
    - Clear role definition and responsibility distribution required
    - Regular team meetings and communication expected
    - Conflict resolution mechanisms should be established early
- Faculty Mentor Assignment:
    - Each team will be assigned a faculty mentor for guidance
    - Mentors provide technical guidance, progress monitoring, and feedback
    - Mentors conduct bi-weekly reviews throughout the semester
    - Students must maintain regular communication with their mentor

4.2 Topic Selection

Alignment Requirements:

- Topics must align with B.Tech program outcomes
- Focus on problem-solving, innovation, and technical proficiency
- Projects should address real-world challenges in areas such as given in the list below (but not limited to):
    - Software Development and Applications

- Embedded Systems and IoT
- Artificial Intelligence and Machine Learning
- Data Science and Analytics
- Cybersecurity
- Cloud Computing and Distributed Systems
- Sustainable Engineering and Green Technology
- Robotics and Automation
- Mobile and Web Technologies
- Network Systems

Approval Process:

1. **Proposal Submission:** Submit a 1-page project proposal (maximum 500 words) including:
    - Proposed project title
    - Clear problem statement
    - Project objectives (3-5 SMART objectives)
    - Proposed methodology and tools
    - Expected outcomes and deliverables
    - Timeline overview
2. **Submission Timeline:** Beginning of the semester (Week 2)
3. **Approval Authority:** Faculty Mentor and Academics Team
4. **Review Criteria:**
    - Feasibility within semester timeframe
    - Technical challenge level appropriate for 7th semester
    - Innovation potential
    - Resource availability
    - Industry or societal relevance
    - Alignment with program outcomes
5. **Feedback and Revision:** Teams may be asked to revise proposals based on faculty feedback

Examples of Suitable Topics:

- Smart irrigation system using IoT and machine learning for precision agriculture
- Real-time traffic management system with edge computing capabilities
- Renewable energy optimization platform for microgrids
- AI-powered healthcare diagnosis system using deep learning
- Blockchain-based supply chain transparency solution

- Intelligent home automation system with voice and gesture control
- Cybersecurity threat detection using anomaly-based algorithms
- E-waste management and recycling tracking system
- Augmented reality application for education and training
- Predictive maintenance system for industrial equipment.

5 PRESENTATIONS AND EVALUATION

5.1 Evaluation Schedule

Three formal evaluations will be conducted during the semester to assess project progress and quality:

Evaluation	Week	Weightage	Focus Areas
Initial Proposal Review	Week 4	20%	• Topic feasibility and relevance • Problem statement clarity. • Objectives and scope definition • Preliminary technology review • Project plan and timeline
Mid-Term Review	Week 8	30%	• Progress assessment. • Methodology implementation • Preliminary results or prototypes • Technology review completion • Identification of challenges • Revised timeline if needed
Final Evaluation	Week 12	50%	• Complete project outcomes • Working prototype/system demonstration • Report quality and completeness • Presentation effectiveness • Innovation and impact assessment • Objective achievement

Important Notes:

- Attendance at all evaluations is mandatory
- Teams must submit required documents 48 hours before each evaluation

- Failure to attend an evaluation without valid reason may result in zero marks for that evaluation
- Medical or emergency absences must be documented and approved by the Program Director.

5.2 Evaluation Format

Presentation Structure:

- Duration: 15 minutes presentation + 5 minutes Q&A per team
- Format: PowerPoint or PDF presentation with live demonstration (if applicable)
- All Team Members: Must participate in the presentation

Evaluation Panel:

- Faculty Mentor (primary evaluator)
- Program Director or Department Head
- External industry expert or academic specialist (for final evaluation)

Evaluation Rubrics:

Criterion	Weightage	Description
Problem Identification	20%	Clarity and relevance of problem statement, understanding of context, scope definition
Methodology	25%	Appropriateness of approach, technical soundness, use of tools and technologies, implementation quality
Innovation and Industry Relevance	25%	Novelty of solution, gap addressing, technical advancement, practical applicability, societal impact
Presentation Clarity	15%	Organization and structure, visual quality, delivery confidence, time management, Q&A handling

Report Quality	15%	Formatting adherence, writing quality, completeness, technical accuracy, plagiarism compliance

Performance Levels:

- **Excellent (90-100%):** Exceeds expectations in all aspects
- **Good (75-89%):** Meets expectations with minor areas for improvement
- **Satisfactory (60-74%):** Acceptable work with significant areas needing improvement

Unsatisfactory (<60%): **Does not meet minimum standards.**

5.3 Tips for Best Presentation

Slide Design:

- Use consistent formatting throughout (font size 24 for headings, 18 for body text)
- Maintain a professional color scheme (avoid overly bright or clashing colors)
- Avoid cluttered slides - use the "one idea per slide" principle
- Use high-quality images and diagrams
- Include slide numbers for easy reference during Q&A
- Use bullet points sparingly - prefer visuals over text
- Ensure text is readable from a distance

Content Organization:

- Start with a compelling title slide
- Include an agenda/outline slide
- Focus on key points: problem, methodology, results, impact
- Use visuals effectively (diagrams, charts, graphs, screenshots)
- Include live demonstration or video if possible
- Show comparisons with existing solutions
- Highlight novelty and contributions clearly
- End with a strong conclusion and future work slide

Delivery Excellence:

- Practice multiple times before the presentation (minimum 3-4 rehearsals)
- Speak clearly, confidently, and at an appropriate pace
- Make eye contact with panel members
- Use natural gestures and body language
- Avoid reading directly from slides
- Engage the audience with questions or demonstrations
- Distribute presentation time fairly among team members
- Prepare for technical difficulties with backup plans

Q&A Preparation:

- Anticipate potential questions and prepare answers
- Listen carefully to each question before answering
- Answer concisely and directly
- If unsure, acknowledge it honestly and offer to follow up
- Refer to slides or documentation when needed
- Remain calm and professional even with challenging questions
- All team members should be prepared to answer questions

Time Management:

- Strictly adhere to the 15-minute presentation limit
- Allocate time appropriately across sections
- Keep a timer visible during practice sessions
- Prepare a "quick version" in case you need to speed up
- Reserve 1-2 minutes for conclusion and questions transition

6 FINAL DOCUMENTATION AND SUBMISSION

Submission Deadline: End of Week 12 (specific date to be announced at the beginning of the semester)

Submission Requirements:

1. **Project Report:**
    - Format: PDF file (not editable formats like .docx)
    - File naming convention: CapstoneI_[TeamID]_[ProjectTitle].pdf
    - File size: Maximum 50 MB
    - Must adhere to all formatting guidelines specified in Section 3
2. **Presentation Slides:**
    - Format: PDF or PowerPoint (.pptx)
    - Include notes/script if available
3. **Source Code and Documentation:**
    - Complete, well-commented source code
    - README file with setup instructions
    - User manual or documentation
    - Format: ZIP file with organized folder structure
4. **Demonstration Video (Optional but Recommended):**
    - Duration: 5-10 minutes
    - Shows working prototype/system in action
    - Format: MP4 (maximum 200 MB)
5. **Project Submission Form:** (See Annexure 1)
6. **Plagiarism Report:**
    - Official plagiarism check report from institutional tool
    - Must show similarity index <15%

Submission Method:

- All materials must be uploaded to the designated Learning Management System (LMS) portal
- One hard copy of the report must be submitted to the department office
- Hard copy should be spiral-bound with a transparent front cover

Vetting Process:

- Final reports will be reviewed by the Evaluation Panel and Academics Team
- Review for compliance with:
    - Academic standards and formatting guidelines
    - NBA/NAAC accreditation requirements
    - Plagiarism policies
    - Technical completeness
- Teams may be asked to make revisions if deficiencies are identified
- Final grade release only after successful vetting

Archiving:

- All project records will be archived for a minimum of 5 years for:
  - Accreditation and quality assurance purposes
  - Future reference and research
  - Institutional knowledge repository
- Outstanding projects may be showcased in:
  - University project exhibition
  - Department website
  - Industry interaction sessions

7 SUMMARY

7.1 Project Timeline Overview

Week	Activity	Milestone	Deliverable
Week 1-2	Team formation and topic selection	Team finalization	Team composition form, Project proposal
Week 3	Topic approval and project planning	Approval received	Detailed project plan
Week 4	Initial Proposal Review	Evaluation 1 (20%)	Proposal presentation
Week 5-7	Technology review, design, and initial implementation	Design completion	System architecture, Technology review table
Week 8	Mid-Term Review	Evaluation 2 (30%)	Progress presentation, Preliminary prototype

Week 9-11	Implementation, testing, and refinement	System integration	Working prototype, Test results
Week 12	Final Evaluation	Evaluation 3 (50%)	Final presentation, Complete report

7.2 Evaluation Criteria and Weightage

Evaluation Criterion	Weightage (%)	Key Assessment Points
Problem Identification	**20%**	**Clarity, relevance, real-world significance, scope definition**
Methodology	**25%**	**Technical approach, tool selection, implementation quality, systematic process**
Innovation and Industry Relevance	**25%**	**Novelty, gap addressing, technical advancement, practical impact**
Presentation Clarity	**15%**	**Organization, visual quality, delivery, time management, Q&A**

Report Quality	**15%**	**Formatting, writing, completeness, technical accuracy, originality**
	TOTAL	100%

7.3 Grade Distribution

Grade	Marks Range	Performance Level
O (Outstanding)	90-100	Exceptional work exceeding all expectations
A+ (Excellent)	80-89	Excellent work meeting all criteria with distinction
A (Very Good)	70-79	Very good work with minor areas for improvement
B+ (Good)	60-69	Good work meeting basic requirements
B (Above Average)	50-59	Satisfactory work with several areas needing improvement
C (Average)	40-49	Acceptable work meeting minimum standards

F (Fail)	<40	Unsatisfactory work not meeting minimum standards

7.4 Important Compliance Notes

Plagiarism Policy:

- Maximum similarity index allowed: <10%
- All sources must be properly cited using IEEE format
- Direct copying without attribution constitutes academic misconduct
- Violations may result in project disqualification and disciplinary action

Academic Ethics:

- All work must be original and completed by team members
- External help must be acknowledged in the report
- Data fabrication or falsification is strictly prohibited
- Proper ethical clearance required for projects involving human subjects or sensitive data

8 ANNEXURES:

8.1 Annexure – I: PROJECT SUBMISSION FORM

Capstone Project-I Submission Form

B.Tech 7th Semester, Academic Year 2025-2026

Project Title:	
Department:	
Team ID:	

Team Members:

S.No.	Name	Roll No.	Signature
1.			
2.			
3.			

Faculty Mentor:	
Date of Submission:	

Submission Checklist:

- ☐ **Project Report (PDF format)**
- ☐ **Presentation Slides**
- ☐ **Source Code and Documentation (ZIP file)**
- ☐ **Plagiarism Report (<15% similarity)**
- ☐ **Project Log/Work Diary**
- ☐ **Demonstration Video (if applicable)**
- ☐ **Hard Copy of Report (Spiral Bound)**

Faculty Mentor Signature Department Seal

8.2 ANNEXURE - II: SAMPLE PROJECT STRUCTURE

The following structure provides a chapter-wise breakdown of expected content for the Capstone Project-I report. Use this as a guide to organize your report systematically.

8.2.1 Preliminary Pages

- **Title Page**
- **Table of Contents**
- **List of Figures (if applicable)**
- **List of Tables (if applicable)**
- **List of Abbreviations (if applicable)**

8.2.2 Chapter 1: Introduction (10-12% of total word count)

- **1.1 Background:** Context and domain overview (2-3 pages)
- **1.2 Problem Statement:** Clear articulation of the problem (1 page)
- **1.3 Significance:** Why this project matters (1 page)
- **1.4 Scope:** Boundaries and limitations (1 page)
- **1.5 Organization of Report:** Brief overview of remaining chapters (0.5 page)

8.2.3 Chapter 2: Technology Review (20-25% of total word count)

- **2.1 Introduction:** Overview of review methodology (0.5 page)
- **2.2 Related Technologies:** Detailed review using structured table (4-5 pages)
- **2.3 Comparison and Analysis:** Comparative analysis of technologies (2-3 pages)
- **2.4 Review Summary:** Key findings and insights (1-2 pages)
- **2.5 Research Gap:** Identified limitations and opportunities (1-2 pages)

8.2.4 Chapter 3: Objectives and Methodology (20-25% of total word count)

- **3.1 Project Objectives:** SMART objectives (1 page)
- **3.2 Proposed Approach:** Overall strategy and methodology (2-3 pages)
- **3.3 System Architecture:** Design and components with diagrams (2-3 pages)
- **3.4 Tools and Technologies:** Detailed description of selected tools (2 pages)
- **3.5 Implementation Plan:** Phases and timeline with Gantt chart (1-2 pages)
- **3.6 Novelty:** Unique contributions and innovations (1-2 pages)

8.2.5 Chapter 4: Implementation and Results (25-30% of total word count)

- **4.1 Implementation Details:** Step-by-step implementation process (3-4 pages)
- **4.2 System Features:** Detailed description of implemented features (2-3 pages)
- **4.3 Testing and Validation:** Testing methodology and results (2-3 pages)
- **4.4 Results and Analysis:** Outcomes with data, graphs, screenshots (3-4 pages)
- **4.5 Discussion:** Interpretation and implications (2 pages)

8.2.6 Chapter 5: Conclusion and Future Work (10-12% of total word count)

- **5.1 Summary:** Recap of project and achievements (1 page)
- **5.2 Objectives Achievement:** How objectives were met (1 page)
- **5.3 Limitations:** Current constraints and challenges (1 page)
- **5.4 Lessons Learned:** Key takeaways (1 page)
- **5.5 Recommendations:** Suggestions for improvement (1 page)
- **5.6 Future Scope:** Potential extensions for Capstone-II (1-2 pages)

8.2.7 End Matter

- Acknowledgment
- References (IEEE format, minimum 15-20 references)
- Appendices (if any):
    - Appendix A: Source Code Snippets (key algorithms only)
    - Appendix B: User Manual/Documentation
    - Appendix C: Survey Forms/Questionnaires (if applicable)
    - Appendix D: Additional Data/Results

Importan Notes:

- This structure is a guideline; adapt based on your specific project needs
- Maintain consistent numbering and formatting throughout
- Ensure smooth transitions between chapters
- Cross-reference figures, tables, and sections appropriately
- Total word count: 3,000-4,000 words (excluding references and appendices)

8.3 ANNEXURE - III: EVALUATION SHEET

INTERNAL EVALUATION SHEET - CAPSTONE PROJECT-I

Project Title: _____

Team Members: _____

Faculty Mentor: _____

Evaluation Type: ☐ **Initial Proposal (20%)** ☐ **Mid-Term (30%)** ☐ **Final (50%)**

Date: _____

S.No.	Criterion	Weightage (%)	Marks Awarded (out of 100)	Weighted Score
1	**Problem Identification**	**20**		
2	**Methodology**	**25**		
3	**Innovation and Industry Relevance**	**25**		
4	**Presentation Clarity**	**15**		
5	**Report Quality**	**15**		
	TOTAL	100		

Comments and Feedback:
Recommendations for Improvement:

Evaluator 1 Signature Evaluator 2 Signature